

UPPER TAIL BOUNDS FOR IRREGULAR GRAPHS

ANIRBAN BASAK AND SHAIBAL KARMAKAR

ABSTRACT. In this note we consider the upper tail large deviations of subgraph counts for irregular graphs H in $\mathbb{G}(n, p)$, the sparse Erdős-Rényi graph on n vertices with edge connectivity probability $p \in (0, 1)$. For $n^{-1/\Delta} \ll p \ll 1$, where Δ is the maximum degree of H , we derive the upper tail large deviations for any irregular graph H . On the other hand, we show that for p such that $1 \ll n^{v_H} p^{e_H} \ll (\log n)^{\alpha_H^*/(\alpha_H^*-1)}$, where v_H and e_H denote the number of vertices and edges of H , and α_H^* denotes the fractional independence number, the upper tail large deviations of the number of unlabelled copies of H in $\mathbb{G}(n, p)$ is given by that of a sequence of Poisson random variables with diverging mean. Restricting to the r -armed star graph we further prove a localized behavior in the intermediate range of p (left open by the above two results) and show that the mean-field approximation is asymptotically tight for the logarithm of the upper tail probability.

1. INTRODUCTION AND MAIN RESULTS

The classical large deviation theory traditionally deals with large deviations of linear functions of independent and identically distributed (i.i.d.) random variables. Probably the simplest non-trivial problem that falls outside realm of the classical theory is the large deviations of triangle counts in an Erdős-Rényi random graph, to be denoted by $\mathbb{G}(n, p)$, with vertex set $\llbracket n \rrbracket := \{1, 2, \dots, n\}$ and edge connectivity probability $p = p_n \in (0, 1)$, where each pair of vertices is connected with a probability p independently of every other pair. In the last fifteen years there have been immense interest in studying large deviations subgraph (in particular triangle) counts in Erdős-Rényi random graphs, both in the dense regime, i.e. $p \asymp 1$ (see the monograph [9] and the references therein), and in the sparse regime, i.e. $p \ll 1$, e.g. see [2, 7, 10, 12, 13, 14] (we refer the reader to Section 1.5 for all the notation).

Recently Harel, Mousset, and Samotij [15] showed that the speed and the rate function of upper tail of cliques (i.e. complete subgraphs) on r vertices in $\mathbb{G}(n, p)$ undergoes a transition at the threshold $p^{(r-1)/2} \asymp n^{-1}(\log n)^{1/(r-2)}$. For $p^{(r-1)/2} \ll n^{-1}(\log n)^{1/(r-2)}$ (and $p^{(r-1)/2} \gg n^{-1}$ so that the expected number of cliques on r vertices diverges) the speed and the rate function are given by those for a sequence of Poisson random variables with diverging means, and thus that regime of p is rightly termed as the *Poisson regime*. In contrast, for $p^{(r-1)/2} n^{-1}(\log n)^{1/(r-2)} \ll p^{(r-1)/2} \ll 1$ the large deviation event is primarily due to the presence of some localized structures (of microscopic size) in the graph, and therefore that regime is termed as the *localized regime*.

It was conjectured in (an earlier version of) the work [15] that the transition between the localized and the Poisson regimes, for the upper tail of *all* connected regular subgraph counts, should occur at the threshold $p^{\Delta/2} \asymp n^{-1}(\log n)^{1/(v_H-2)}$, where Δ is the common degree of the regular graph H in context and v_H denotes the number of vertices in H . In [15] the Poisson behavior was proved in the predicted Poisson regime, and the localized behavior for p such that $n^{-1/2-o(1)} \leq p^{\Delta/2} \ll 1$ (and under the additional assumption that H is non bipartite in the extended range $n^{-1}(\log n)^{\Delta v_H^2} \ll p \ll 1$). The more recent work [5] extends the localized behavior for *all* regular graph in the sub regime left open by [15]. Therefore, [15] and [5] together settle the problem of upper tail large deviations of regular subgraph counts in sparse Erdős-Rényi graphs (albeit a couple of boundary cases). Let us also mention in passing that the speed and the rate function in the localized regime turns out to be the solution of an appropriate mean field variational problem. See Section 1.4 below for a further discussion on this.

In this short article we investigate whether a transition threshold between the localized and the Poisson regimes exists, and whether in the localized regime the large deviation probability can be expressed as the solution of some mean field variational problem for the upper tail of irregular subgraph (i.e. there exists at least two vertices with unequal degrees) counts in sparse Erdős-Rényi graphs.